

Credit Risk Segmentation & NPA Reduction

A portfolio case study on identifying concentrated default risk and turning existing data into practical underwriting and collection decisions.

24,626
applications

13,011
disbursed loans

1.34%
baseline NPA

174
unique defaults

THE CORE QUESTION

Where is credit risk actually coming from — and what can GrayQuest change without adding unnecessary friction to low-risk borrowers?

PROJECT SNAPSHOT

Output	What it delivered
4-variable credit risk scorecard	0–9 score → four operational risk tiers
Logistic regression	Default-risk evidence; ROC-AUC 0.7871
Early Warning System	8 pre/post-disbursement triggers
Six policy simulations	NPA vs revenue trade-off quantified
Implementation roadmap	Month 1–12 operating plan

Portfolio lens: Risk concentration → scoring → early warning → policy simulation → action.

The portfolio average was only part of the story.

GrayQuest's 1.34% overall NPA rate concealed several concentrated risk pockets. The analysis focused on finding those pockets and determining which ones could be influenced through policy or process.

Risk pocket	Evidence	Business meaning
Manual approval	2.36% vs 0.59% bureau	4× higher risk; 74.7% of NPAs
11-EMI	2.45% vs 0.43% 6-EMI	5.7× higher rate; 54.6% of defaults
Ratnaafin Lite	5.15% NPA	Highest partner rate; concentrated exposure
30–60 DPD	100 / 181 NPA accounts	₹59.95L still in recovery window

PROJECT OBJECTIVE

Build a framework that could score applicants before disbursement, flag emerging risk before NPA crystallisation, simulate approval policies, and support a target below 1% NPA.

2 institutions	6 NBFC partners	181 NPA records	Nov 2025 NACH cycle
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Included	Excluded
13,011 disbursed FY 2024–25 loans; Sri Chaitanya + Narayana; six NBFC partners; borrower, approval, tenure, bureau and collection signals.	Pre-credit marketing funnel; NBFC internal operations; revenue / P&L; modelling.

THE BUSINESS QUESTION

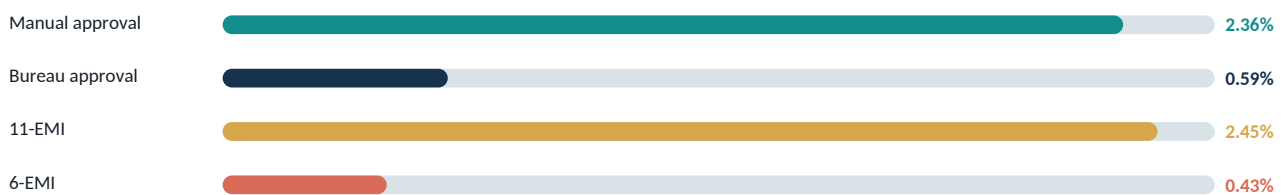
Which borrower and process signals can be converted into earlier intervention — without blanket tightening of the entire portfolio?

Five analytical moves. One decision framework.

The work progressed from data preparation to operationalisation; the model was an input, not the endpoint.

Stage	Work	Output
01 • Prepare	24,626 applications; 4,541 duplicates identified; 13,011 disbursed loans isolated.	Analysis-ready master
02 • Diagnose	CIBIL, income, tenure, approval channel, NBFC and profile segmentation.	Risk concentration map
03 • Validate	Welch t-test, Z-test, ANOVA, χ^2 and correlation analysis.	Evidence-backed drivers
04 • Predict	Binary logistic regression; AUC 0.7871; McFadden R^2 10.67%; threshold 0.055.	Default-risk evidence
05 • Operationalise	Scorecard + 8 EWS triggers + 6 policy simulations.	Decision framework

STRONGEST RAW-RATE GAPS



Signal	Finding	Actionable interpretation
Loan amount	p=0.774	Not a meaningful standalone default driver
CIBIL	Defaulters ~25 points lower	Useful bureau risk discriminator
Approval channel	Manual 4× bureau NPA	Strong policy lever
Tenure	11-EMI ~5.7× 6-EMI NPA	Strong policy lever

ANALYTICAL PRINCIPLE

Prefer interpretable signals that GrayQuest can actually use in underwriting and collections.

Make risk visible before the loan is disbursed.

Four observable dimensions were converted into a simple additive 0–9 score.

Dimension / category	Pts	Observed NPA	Read-through
CIBIL >750	0	0.73%	Lowest-risk band
CIBIL 700–750	1	2.35%	Moderate risk
CIBIL 650–700	2	4.48%	~6× >750 risk
CIBIL NTC	2	2.64%	No bureau history
CIBIL <650	3	Insufficient data	Highest bureau risk
Salaried	0	0.95%	Stable income
Self-Employed	1	2.69%	2.84× salaried
6-EMI	0	0.43%	Lowest tenure risk
9-EMI	1	1.98%	4.6× 6-EMI
11-EMI	2	2.45%	Strong predictor
Bureau / Banking	0	0.59%	Data-driven channel
Manual Override	3	2.36%	4× bureau; ~75% NPAs

MAXIMUM SCORE = 9

CIBIL <650 [3] + Self-Employed [1] + 11-EMI [2] + Manual Approval [3]

OBSERVED SCORE DISTRIBUTION

Score	Portfolio	Loans	Default rate	Tier
0	25.7%	3,338	0.06%	Low
1	13.0%	1,692	0.71%	Low
2	12.1%	1,579	0.95%	Medium
3	13.2%	1,718	0.93%	Medium
4	10.7%	1,396	0.86%	High
5	11.8%	1,531	2.74%	High
6	8.5%	1,110	3.15%	Very High
7	4.1%	538	5.20%	Very High
8	0.8%	109	11.01%	Very High

Turn the score into different levels of friction.

Tier	Score	Portfolio	Decision	Core control
Low	0-1	73.0%	AUTO-APPROVE	Basic KYC; fast-track
Medium	2-3	14.6%	CONDITIONAL	Income proof + bank statement; ₹75k cap
High	4-5	6.5%	ENHANCED REVIEW	Senior sign-off; co-applicant; restrict 11-EMI
Very High	6+	5.9%	REJECT	No override; GQ_Defaulter block

LOGISTIC REGRESSION — SELECTED EVIDENCE

Variable	Odds ratio	Interpretation
CIBIL (1 SD ↑)	0.812	~18.8% lower default odds
Self-Employed	1.995	~2× higher odds
Manual Approval	2.208	~2.2× higher odds
9-EMI	3.896	~3.9× vs 6-EMI
11-EMI	3.967	~4× vs 6-EMI; strongest predictor
Overdue flag	1.858	~1.9× higher odds

0.7871 ROC-AUC	10.67% McFadden R²	0.055 threshold
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EARLY WARNING SYSTEM

Stage	Trigger	Coverage	Action
PRE	CIBIL <700 + overdue	4.0%	Conditional + enhanced docs
PRE	CIBIL <700 + overdue + SE	1.2%	Hard reject / extreme conditions
PRE	Any 90+ DPD	5.8%	Senior review
PRE	Combined score ≥5	4.3%	Reject unless compensating factors
POST	DPD 30+	13.6%	Proactive outreach
POST	1st NACH failure	~7-12%	SMS + call within 24h
POST	2 consecutive failures	Very high	Field visit + payment plan
POST	SC mid-month failure	11.55%	Dedicated collection team

Six ways to reduce NPA. One selected balance.

Each scenario tests the same portfolio against a different approval rule and quantifies the risk/revenue trade-off.

Scenario	Policy	Rejected	Prevented	NPA after	NPA saved	Revenue foregone
A	GQ_Defaulters only	258	19	1.19%	₹10.8L	₹32.6L
B	CIBIL_Defaulters (manual)	992	44	1.00%	₹25.1L	₹115.1L
C	GQ + CIBIL_Defaulters	1,250	63	0.85%	₹35.9L	₹147.7L
D	Score ≥5	992	53	0.93%	₹30.2L	₹118.4L
E	Score ≥4	1,606	73	0.77%	₹41.6L	₹193.1L
F	Score ≥6 + conditional Medium	1,333	57	0.91%	₹32.5L	₹156.4L

SCENARIO F — SELECTED

Reject Very High Risk (Score ≥6) and conditionally approve Medium Risk. It projects 0.91% NPA, prevents ~57 defaults and saves ₹32.5L NPA outstanding while affecting 10.3% of disbursals.

WHY F INSTEAD OF E

Policy	NPA after	Portfolio affected	Revenue foregone
E	0.77%	12.3% rejected	₹193.1L
F	0.91%	10.3% affected	₹156.4L

Scenario E reaches the lowest simulated NPA, but with a larger portfolio and revenue sacrifice. Scenario F was documented as the best balance between NPA reduction and revenue trade-off.

From analysis to an operating model.

Area	Priority action
Underwriting	Mandatory scorecard gate; GQ_Defaulter block; dual sign-off for CIBIL_Defaulter; restrict 11-EMI for higher-risk profiles.
Collections	Same-day response to first NACH failure; field escalation after second failure; institution-specific collection resources.
Product / NBFC	Review Ratnaafin Lite; favour 6-EMI for higher-risk segments; route stronger profiles toward stronger partners.
Data / governance	Mandatory default-reason capture; monthly NPA dashboard; duplicate resolution; quarterly scorecard recalibration.

12-MONTH IMPLEMENTATION

Phase	Timing	Key actions	NPA target
Quick wins	M1-2	GQ_Defaulter block; dual sign-off; same-day NACH response	1.19%
Scorecard	M2-4	Full scorecard; 11-EMI restriction; collection tracks; partner review	1.05%
Full tiering	M4-6	Scenario F; field agents; NPA dashboard; tenure policy	0.91%
Optimise	M6-12	Quarterly recalibration; duplicate resolution; partner routing	<0.85%

1.34% → 0.91% projected NPA	57 defaults prevented	₹32.5L NPA saved / cycle	10.3% disbursals affected
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RESPONSIBLE INTERPRETATION

The analysis covers two institutional partners and the specified FY 2024-25 disbursed portfolio. NACH analysis uses the November 2025 cycle. The regression sample excludes six defaults with missing CIBIL. No train-test split was performed; the model should therefore be treated as a policy/portfolio analytics tool rather than a production-ready predictor.

PORTFOLIO TAKEAWAY

GrayQuest's NPA issue was not uniform. It was concentrated in manual approvals, 11-EMI tenure, repeat-default profiles and the 30-60 DPD window. The strongest response is targeted intervention — faster decisions for low-risk borrowers, stronger friction where risk concentrates, and earlier collection action.

Portfolio-ready summary: Built a data-driven credit risk framework using 24,626 applications and 13,011 disbursed loans; developed a 4-variable scorecard, logistic regression model and 8-trigger EWS; simulated six approval policies and selected Scenario F, projecting NPA improvement from 1.34% to 0.91%.

Source: GrayQuest internship report "Credit Risk Segmentation & NPA Reduction Strategy for GrayQuest Education Finance" and accompanying presentation. Figures retained as documented.